

Inspiring Excellence

COMPUTER NETWORKS

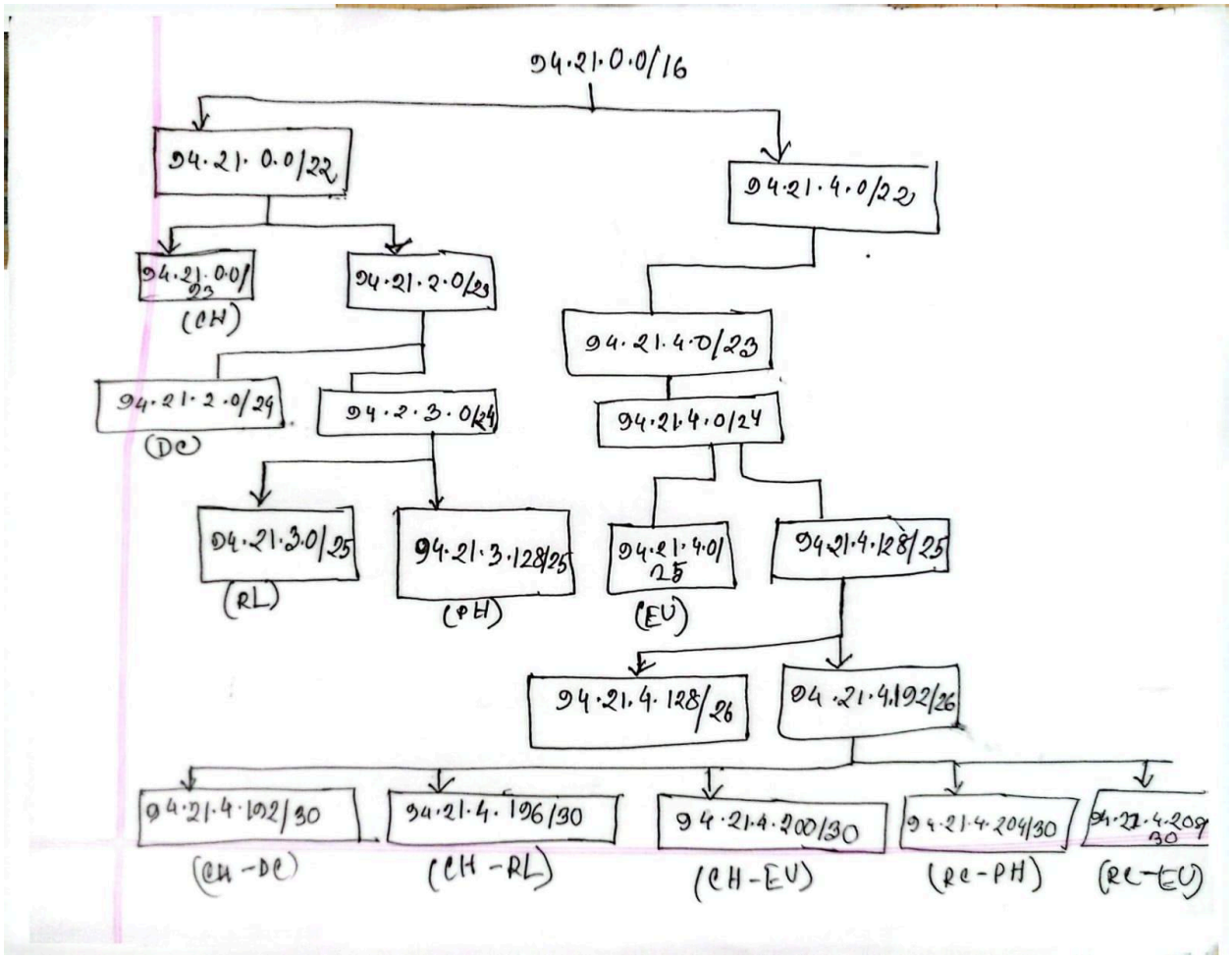
CSE421
Lab Project

Topic 9: Smart Healthcare Network Design

Group Number: 2207

ID	Name
22299421	AUDRIJA CHOWDHURY
22299141	NAFISA HASAN
22299138	SWAPNIL ROB

1. VISM TREE:


94.21.0.0/16
94.21.0.0/22
94.21.4.0/22
94.21.0.0/23
(CH)
94.21.2.0/23
94.21.4.0/23
94.21.2.0/24
(DC)
94.2.3.0/24
94.21.4.0/24
94.21.3.0/25
(RL)
94.21.3.128/25
(PH)
94.21.4.0/25
(EU)
94.21.4.128/25
94.21.4.128/26
94.21.4.192/26
94.21.4.192/30
(CH-DC)
94.21.4.196/30
(CH-RL)
94.21.4.200/30
(CH-EU)
94.21.4.204/30
(RL-PH)
94.21.4.208/30
(RL-EU)

2. IP Addressing Table:

2.1 For routers:

Unit	Network Address	Subnet Mask	Prefix	Usable IP Range	Broadcast Address
CH	94.21.0.0	255.255.254.0	/23	94.21.0.1 – 94.21.1.254	94.21.1.255
DC	94.21.2.0	255.255.255.0	/24	94.21.2.1 – 94.21.2.254	94.21.2.255
RL	94.21.3.0	255.255.255.128	/25	94.21.3.1 – 94.21.3.126	94.21.3.127
PH	94.21.3.128	255.255.255.128	/25	94.21.3.129 – 94.21.3.254	94.21.3.255
EU	94.21.4.0	255.255.255.128	/25	94.21.4.1 – 94.21.4.126	94.21.4.127
RC	94.21.4.128	255.255.255.192	/26	94.21.4.129 – 94.21.4.190	94.21.4.191

2.2 Router Interfaces:

Device	Interface	IP Address	Subnet Mask	Connected To
CH	G0/0	94.21.0.1	255.255.254.0	CH LAN (Switch0)
	S0/0/0	94.21.5.1	255.255.255.252	DC
	S0/0/1	94.21.5.5	255.255.255.252	RL
	S0/1/0	94.21.5.9	255.255.255.252	EU
DC	G0/0	94.21.2.1	255.255.255.0	DC LAN (Switch1)
	S0/0/0	94.21.5.2	255.255.255.252	CH
RL	G0/0	94.21.3.1	255.255.255.128	RL LAN (Switch2)
	G0/1	94.21.5.25	255.255.255.248	Core Switch
	S0/0/0	94.21.5.6	255.255.255.252	CH
PH	G0/0	94.21.3.129	255.255.255.128	PH LAN (Switch3)
	G0/1	94.21.5.26	255.255.255.248	Core Switch
	S0/0/0	94.21.5.14	255.255.255.252	RC
EU	G0/0	94.21.4.1	255.255.255.128	EU LAN (Switch5)
	G0/1	94.21.5.27	255.255.255.248	Core Switch
	S0/0/0	94.21.5.10	255.255.255.252	CH

	S0/0/1	94.21.5.18	255.255.255.252	RC
RC	G0/0	94.21.4.129	255.255.255.192	RC LAN (Switch4)
	S0/0/0	94.21.5.13	255.255.255.252	PH
	S0/0/1	94.21.5.17	255.255.255.252	EU

2.3 Server IP Address Table:

Server	IP Address	Subnet Mask	Default Gateway
CH DNS Server	94.21.0.2	255.255.254.0	94.21.0.1
CH Web Server	94.21.0.3	255.255.254.0	94.21.0.1
CH Email Server	94.21.0.4	255.255.254.0	94.21.0.1
DC Email Server	94.21.2.2	255.255.255.0	94.21.2.1
RL Email Server	94.21.3.3	255.255.255.128	94.21.3.1
RL DHCP Server	94.21.3.2	255.255.255.128	94.21.3.1
PH Email Server	94.21.3.131	255.255.255.128	94.21.3.129
PH DHCP Server	94.21.3.130	255.255.255.128	94.21.3.129
EU Email Server	94.21.4.2	255.255.255.128	94.21.4.1
RC Email Server	94.21.4.130	255.255.255.192	94.21.4.129

3.Router configuration commands:

3.1 CH configuration:

```
enable
config t

interface g0/0
 ip address 94.21.0.1 255.255.254.0
 no shutdown
exit

interface s0/0/0
 ip address 94.21.5.1 255.255.255.252
 clock rate 64000
 no shutdown
exit

interface s0/0/1
 ip address 94.21.5.5 255.255.255.252
 clock rate 64000
 no shutdown
exit

interface s0/1/0
 ip address 94.21.5.9 255.255.255.252
 clock rate 64000
 no shutdown
exit

ip route 94.21.2.0 255.255.255.0 94.21.5.2

router rip
 version 2
 network 94.0.0.0
 no auto-summary
 redistribute static
exit

end
```

3.2 DC Router Configuration:

```
en
config t

interface g0/0
 ip address 94.21.2.1 255.255.255.0
 no shutdown
exit

interface s0/0/0
 ip address 94.21.5.2 255.255.255.252
 no shutdown
exit

ip route 0.0.0.0 0.0.0.0 94.21.5.1

ip dhcp excluded-address 94.21.2.1 94.21.2.20
ip dhcp pool DC_POOL
 network 94.21.2.0 255.255.255.0
 default-router 94.21.2.1
 dns-server 94.21.0.2
exit

end
```

3.3 RL configuration:

```
en
config t

interface g0/0
 ip address 94.21.3.1 255.255.255.128
 no shutdown
exit

interface g0/1
 ip address 94.21.5.25 255.255.255.248
 no shutdown
exit
```

```
interface s0/0/0
 ip address 94.21.5.6 255.255.255.252
 no shutdown
exit

router rip
 version 2
 network 94.0.0.0
 no auto-summary
exit

End
```

3.4 PH configuration:

```
en
config t

interface g0/0
 ip address 94.21.3.129 255.255.255.128
 no shutdown
exit

interface g0/1
 ip address 94.21.5.26 255.255.255.248
 no shutdown
exit

interface s0/0/0
 ip address 94.21.5.14 255.255.255.252
 no shutdown
exit

ip route 94.21.2.0 255.255.255.0 94.21.5.25

router rip
 version 2
 network 94.0.0.0
 no auto-summary
```

```
exit

end
```

3.5 EU configuration:

```
en
config t

interface g0/0
 ip address 94.21.4.1 255.255.255.128
 no shutdown
exit

interface g0/1
 ip address 94.21.5.27 255.255.255.248
 no shutdown
exit

interface s0/0/0
 ip address 94.21.5.10 255.255.255.252
 no shutdown
exit

interface s0/0/1
 ip address 94.21.5.18 255.255.255.252
 no shutdown
exit

ip route 94.21.0.0 255.255.254.0 s0/0/0

ip dhcp excluded-address 94.21.4.1 94.21.4.20
ip dhcp pool EU_POOL
 network 94.21.4.0 255.255.255.128
 default-router 94.21.4.1
 dns-server 94.21.0.2
exit

router rip
```

```
 version 2
 network 94.0.0.0
 no auto-summary
 passive-interface s0/0/1
exit

end
```

3.6 RC configuration:

```
en
config t

interface g0/0
 ip address 94.21.4.129 255.255.255.192
 no shutdown
exit

interface s0/0/0
 ip address 94.21.5.13 255.255.255.252
 clock rate 64000
 no shutdown
exit

interface s0/0/1
 ip address 94.21.5.17 255.255.255.252
 clock rate 64000
 no shutdown
exit

ip route 94.21.0.0 255.255.254.0 94.21.5.18 130

router rip
 version 2
 network 94.0.0.0
 no auto-summary
exit

end
```

4. Assumptions:

1. All network devices are functioning properly without hardware failure.
2. Static IP is used for servers and CH/RC networks as required.
3. RIPv2 convergence happens without delay or packet loss.
4. DHCP servers assign IPs correctly to all client devices.
5. Packet Tracer limitations are considered for email communication.